```
clc;
Clear all;
Close all;
D = input ('enter the decimating factor:');
N = input ('enter the no. of samples:');
n = 0:N-1;
x = Sin(2* pi * 0.03 * n);
y = decimate (x,D);
Subplot (2,1,1);
stem( n,x(1:N));
xlabel ('time index');
ylabel( 'amp');
title (' input sequence');
Subplot(2,1,2);
m = 0:(N/D) -1);
xlabel( 'time index');
ylabel('amp');
title ('output sequence');
```

3. Type the program

4. Save in current directory

5. Compile and Run the program

6. For the output see command window\ Figure window

MATLAB PROGRAM

```
clc;
close all;
clear all;
I = input ('enter interpolation factor:');
N = input ('enter no. of input samples:');
n=0:N-1;
x = sin(2*pi*0.03*n);
y = interp(x, I);
subplot (211);
stem(n,x(1:N));
xlabel ('time');
ylabel ('ampl');
title ('input sequence');
subplot(212);
m:0:((N*I)-1);
xlabel('time');
ylabel('amp');
title ('output sequ
```


O/P :-

enter interpolation factor: 2
enter no. of input samples: 20

4. Save in current directory

5. Compile and Run the program

6. For the output see command window\ Figure window

MATLAB PROGRAM

```
clc;
close all;
clear all;
I = input ('enter the interpolation factor');
D = input ('enter the decimation factor');
N = input ('enter the no. of i/p samples');
```

```
n = 0:N-1;
x = sin (2* pi * 0.03 * n) + sin(2 * pi * 0.43 * n)
y = resample ( x, I, D)
subplot (2, 1,1)
stem(n,x)
xlabel( 'time')
ylabel ('amplitude')
title ('i/p sequence')
subplot (2,1,2)
m = 0: (N* I/D) -1
stem (m, y (1: N * I/D))
xlabel ('time')
ylabel ('amplitude')
title ('o/p sequence')
```